

2(a)(i)	1 description of random / systematic, sampling ; 2 use of, quadrats / (sweep) nets / (light) trap / pooter ; 3 Count the number of each phenotypes ; OR max. 2 marks if Mark-Release-Recapture method 4 <u>capture</u> ladybirds and count / find number / record, number of each phenotypes and <u>mark and release</u> them ; 5 suitable marking so not harmful / too obvious / not removed 6 <u>capture</u> second sample and count / find number / record, number of each phenotypes and count / find number / record number, of marked each phenotypes ;	3
2(a)(ii)	any two from: 1 <i>ref. to</i> data for spring 1929 and spring 1932 ; 2 sample size(s) ; 3 multiple sampling sites (in same area of Germany) ; 4 carry out statistical analysis ; <i>Sampling took place with:</i> 5 the same location(s) / habitat(s) / sampling site(s) / AW ; 6 the same time of day ; 7 the same duration (of sampling) ;	2
2(b)(i)	Line for melanic phenotype is <u>steeper</u> (than non-melanic phenotype) ;	1

Question	Answer	Marks
2(b)(ii)	1 award if both graphs have been used ; any two from: 2 all ladybirds have the same starting temperature ; 3 all ladybirds warm up when lamp is on ; 4 all ladybirds return to, original / starting, temperature ; 5 melanic reaches higher temperature / ora ; 6 melanic cool down faster / ora ; 7 females warm up quicker / ora ; 8 females reach higher temperature / ora ; 9 females take longer to cool down / ora ; 10 heavier warm up quicker / ora ; 11 heavier ladybirds reach higher temperature / ora ; 12 heavier ladybirds take longer to cool down / ora ; 13 the temperature plateaus for all except non-melanic females ;	3
2(c)	1 Pearson's (linear correlation) / Spearman's (rank correlation) ; any two from: <i>Pearson's</i> 2 data (for temperature and mass) is, paired / linked ; 3 data is continuous ; 4 scatter diagram, suggests a <u>linear</u>, relationship / correlation / association ; 5 there are at least 5 (paired) observations ; OR <i>Spearman's</i> 6 data (for temperature and mass) is, paired / linked ; 7 the data is ordinal / interval ; 8 scatter diagram, suggests an <u>increasing</u>, relationship / correlation / association ; 9 there are more than 5 (paired) observations ;	3

Question	Answer	Marks
2(d)	1 cross <u>melanic</u> with non-melanic ; 2 homozygous (dominant) melanic parent(s) will only have melanic offspring and heterozygous (melanic) parent(s) will have both phenotypes in offspring ; <i>Details about breeding experiment – max 1</i> 3 many breeding pairs ; 4 each breeding pair kept separate from other pairs or separate offspring from parents ; 5 count, the different phenotypes of offspring ;	3